

Recommendations Report:

Technical Workshop on Pinniped Predation on Salmon at the Ballard Locks



Photo courtesy of Eric Warner, Muckleshoot Indian Tribe

January 2024

Recommendations Report:

Technical Workshop on

Pinniped Predation on Salmon at the Ballard Locks

January 2024

Prepared by:

Jason Mulvihill-Kuntz and Lauren Urgenson, Lake Washington/Cedar/Sammamish Watershed
(WRIA 8) Salmon Recovery Council

Joe Scordino, retired NOAA/National Marine Fisheries Service

Environmental Science Associates

Funding and Support by:

WRIA 8 Salmon Recovery Council

Any opinions, findings, conclusions, or recommendations expressed in this publication do not necessarily reflect the views of the organization or agency that provided support for the project.

Workshop Participants, Organized by Affiliation

Columbia River Inter-Tribal Fish Commission	Doug Hatch
Makah Tribe	Liz Allyn Jonathan Scordino
Muckleshoot Indian Tribe	Ava Fuller Mike Mahovich Colton Mahovich Eric Warner
NOAA/National Marine Fisheries Service	Robert Anderson
Oregon Department of Fish and Wildlife	Michael Brown Bryan Wright
Suquamish Tribe	Casey Schmidt
U.S. Army Corps of Engineers	David Chapman, Seattle District Katherine Cousins, Seattle District Laurie Ebner, Portland District Vanessa Pepi, Seattle District Shay Perryman, Seattle District Sean Tackley, Northwestern Division Kyle Tidwell, Portland District Kaitlin Whitlock, Seattle District
Washington Department of Fish and Wildlife	Aaron Bosworth Casey Clark Sarah Colosimo Stewart Reinbold
Lake Washington/Cedar/Sammamish Watershed (WRIA 8) Salmon Recovery Council	Jason Mulvihill-Kuntz Lauren Urgenson
Retired	Bob DeLong, former NOAA/NMFS Marine Mammal Laboratory Patrick Gearin, former NOAA/NMFS Marine Mammal Laboratory Steve Jeffries, former WDFW Brent Norberg, former NOAA/NMFS West Coast Regional Office Joe Scordino, former NOAA/NMFS Northwest Regional Office
Environmental Science Associates	Peter Carr Mike Higgins Susan O'Neil

CONTENTS

Contents	i
Acronyms and Abbreviations	ii
SUMMARY OF WORKSHOP RECOMMENDATIONS	iii
I. INTRODUCTION	1
A. Background	1
B. Workshop.....	2
II. REVIEW OF PINNIPED CONTROL MEASURES AT REGIONAL IMPOUNDMENTS	4
A. Chronology of Pinniped Predation and Management Efforts at the Ballard Locks	4
B. Chronology of Pinniped Predation and Management Efforts at Bonneville Dam	8
C. Chronology of Pinniped Predation and Management Efforts at Willamette Falls	10
III. RECOMMENDATIONS FOR PINNIPED MANAGEMENT AT THE BALLARD LOCKS	14
A. Lessons Learned and Recommendations – Pinniped Removal.....	15
B. Lessons Learned and Recommendations – Pinniped Deterrence	15
C. Lessons Learned and Recommendations – Structural and Operational Changes at the Ballard Locks	16
D. Lessons Learned and Recommendations – Comprehensive Monitoring	17
IV. REFERENCES AND ADDITIONAL SOURCE MATERIAL	18
APPENDIX A. Detailed Description of Pinniped Predation Control Measures Used at Impoundments and Their Limitations	25
APPENDIX B. Figures Showing Area Between the Salmon Bay Railroad Bridge and the Ballard Locks	30

ACRONYMS AND ABBREVIATIONS

ADD	acoustic deterrence device
AHD	acoustic harassment device
CRITFC	Columbia River Inter-Tribal Fish Commission
dB	decibel(s)
ESA	Endangered Species Act
IDFG	Idaho Department of Fish and Game
kHz	kilohertz
MMPA	Marine Mammal Protection Act
NMFS	National Marine Fisheries Service
NOAA	National Oceanic and Atmospheric Administration
ODFW	Oregon Department of Fish and Wildlife
SLED	sea lion exclusion device
TAST	Targeted Acoustic Startle Technology
USACE	U.S. Army Corps of Engineers
USDA	U.S. Department of Agriculture
WDFW	Washington Department of Fish and Wildlife
WDW	Washington Department of Wildlife
WRIA	Water Resource Inventory Area
WSAS	Washington State Academy of Sciences

SUMMARY OF WORKSHOP RECOMMENDATIONS

A technical workshop was convened to develop recommendations for reducing/eliminating the effects of harbor seal predation on threatened and dwindling salmon populations at the Hiram M. Chittenden Locks (Ballard Locks). The urgency for addressing harbor seal presence and predation at the Ballard Locks was driven by the dramatic decline in the Lake Washington sockeye salmon population and Endangered Species Act (ESA)-listed Chinook salmon populations remaining far below recovery goals. Although many forums and technical committees are working on addressing ESA-listed salmon recovery and enhancing the sockeye and coho populations, none are directly addressing the pinniped predation problem in and around the Ballard Locks.

Thus, this technical workshop focused on immediate management actions necessary to reduce/eliminate pinniped predation to prevent further decline of these salmon populations as they migrate through the Ballard Locks. Workshop participants drew on lessons learned from pinniped predation control programs at Bonneville Dam, Willamette Falls, and at the Ballard Locks in the 1980–90s to develop the recommendations. The workshop concluded that harbor seal predation observations in recent years and lessons learned about pinniped predation at impoundments were sufficient, and no further “proof” or studies were necessary, to recommend effective pinniped predation control methods be implemented as soon as possible to prevent further salmon population declines, as happened with the loss of Lake Washington steelhead in the 1990s.

Key recommended management actions, which should be implemented concurrently, are as follows:

- **Pinniped Removal:** Permanently remove pinnipeds that occur in the area between the Salmon Bay Railroad Bridge and the Ballard Locks that are not responsive to deterrence efforts.
- **Pinniped Deterrence:** Implement a multifaceted deterrence program during adult and juvenile salmon migrations to prevent pinnipeds from habituating to foraging at the Ballard Locks (i.e., deter newly arriving and naïve pinnipeds so that they do not need to be permanently removed).
- **Structural and Operational Changes at Ballard Locks Facility:** Design and install a pinniped exclusion structure at the entrance to the fish ladder that prevents pinniped access without affecting salmon passage (similar to the sea lion exclusion devices [i.e., barrier gates] at Bonneville Dam and Willamette Falls). Continue studies on the feasibility of structural and operational modifications to enhance salmon migration through the Ballard Locks.
- **Comprehensive Monitoring:** Implement a comprehensive monitoring program focused on salmon passage; pinniped presence, behavior, and predation; and environmental and operational variables at the Ballard Locks. The data collection regime should be structured to inform, on a real-time basis, the implementation of adaptive management measures to minimize or eliminate the pinniped-salmonid conflict at this site. Monitoring should occur during both adult and smolt migration periods for sockeye, Chinook, and coho salmon.

Additional actions recommended to enable meaningful progress on the above management actions are as follows:

- **Outreach and Education:** Implement a public relations campaign focused on the detrimental effects of pinniped predation on salmon at the Ballard Locks and the urgent need for effective measures to control pinniped predation on salmonids.
- **ESA Consultation on Ballard Locks:** Assess the effects of pinniped foraging on salmon passage survival and evaluate additional physical and operational measures to control it in the next Biological Assessment necessary for ESA consultation on the effects of the Ballard Locks facility and operations on ESA-listed Chinook salmon.

I. INTRODUCTION

A technical workshop on pinniped predation at the Ballard Locks was convened by the Lake Washington/Cedar/Sammamish Watershed (Water Resource Inventory Area [WRIA] 8) Salmon Recovery Council to address the urgent need to assimilate lessons learned from previous and ongoing pinniped predation control methods and to use that knowledge to develop appropriate adaptive approaches to reduce/eliminate the adverse effects of current pinniped foraging on salmon at the Ballard Locks.

A. Background

The Ballard Locks is a regional hotspot for pinniped predation on salmon (WSAS 2022). Anadromous salmonids in the Lake Washington basin must pass through the Ballard Locks twice in their life. First, as juveniles migrating out through the locks facility to Puget Sound. Then again, as adults returning through the locks or fish ladder to spawn upstream in the watershed. Salmon species that spawn and rear in WRIA 8 include Chinook salmon (*Oncorhynchus tshawytscha*), coho salmon (*O. kisutch*), sockeye salmon (*O. nerka*), and previously supported steelhead (*O. mykiss*). When returning salmon reach the Ballard Locks, they are confronted with a migration barrier and an abrupt shift between cool saltwater in Puget Sound and warm freshwater in the Ship Canal. Temperatures in the Ship Canal often exceed thresholds for salmon thermal stress and migration barrier/mortality during summer (Urgenson et al. 2021). In response, salmon can delay migration and/or fall back and recycle downstream several times. This congregation of salmon at the Ballard Locks attracts pinnipeds, resulting in increased salmon vulnerability to stress, injury, and mortality from pinniped foraging.

Predation by pinnipeds—harbor seals (*Phoca vitulina*) and California sea lions (*Zalophus californianus*)—at the Ballard Locks is a longstanding and complex limiting factor to salmon recovery in the Lake Washington basin. Pinnipeds routinely utilize the bottleneck to salmon migration at the Ballard Locks to successfully forage on both adult and juvenile salmonids migrating through the system at different times of year. During the 1980s and 1990s, various pinniped control measures were attempted to keep pinnipeds away from this critical bottleneck to salmonid migration. These intensive management efforts to limit California sea lion predation on Lake Washington steelhead were not successful, and these steelhead are now functionally extirpated (Cram et al. 2018).

Pinniped populations have increased substantially in the coastal waters of Washington State and the Salish Sea since Congress enacted the U.S. Marine Mammal Protection Act (MMPA) in 1972. Current pinniped populations are thought to have surpassed historical, pre-colonization levels as a result of the MMPA protection eliminating human harvests (Carretta et al. 2023). The MMPA has been amended a number of times to allow restricted management of expanding pinniped populations and their impacts on west coast ecosystems, but most of the provisions have been too onerous, cumbersome, and time-consuming to allow effective management of pinniped predation on ESA-listed Chinook along with the other two salmon stocks, sockeye and coho (NMFS 1999, Scordino 2010, Pamplin et al. 2021). One aspect of the MMPA that drew much concern at the workshop was the MMPA requirements for “absolute proof” – prior intense site-specific monitoring that demonstrated significant negative impacts, ineffectiveness of non-lethal removals, and identification of individual pinnipeds that were causing the impacts – before the pinnipeds could be considered for permanent removal from a site.

The most recent MMPA amendments in 2018 did give state and tribal wildlife/fishery managers more flexibility in addressing pinniped predation on ESA-listed salmon and species of concern in the Columbia River, but those new provisions do not apply to Puget Sound.

B. Workshop

A group of subject matter experts on managing pinniped predation, salmon recovery, and/or impoundments (dams/locks) was convened for a technical workshop on February 7, 8, and April 21, 2023. The experts included current and retired employees of the National Oceanic and Atmospheric Administration/National Marine Fisheries Service (NOAA/NMFS), Washington Department of Fish and Wildlife (WDFW), Oregon Department of Fish and Wildlife (ODFW), U.S. Army Corps of Engineers (USACE), Muckleshoot Indian Tribe, Suquamish Tribe, Makah Tribe, Columbia River Inter-Tribal Fish Commission (CRITFC), and WRIA 8 Salmon Recovery Council. The purpose of the workshop was to review available information on pinniped management and make technical recommendations for state/federal/tribal managers and policymakers on an effective and adaptive strategy to reduce/eliminate the effects of current-day pinniped predation on salmon at the Ballard Locks. The workshop did not make recommendations for amendments to the MMPA, but rather left that for state/federal/tribal policymakers to determine as necessary.

At the workshop, Jonathan Scordino, a member of the Washington State Academy of Sciences (WSAS) Scientific Committee, presented the results of the WSAS Scientific Committee deliberations and analyses on increased numbers of pinnipeds in Washington and their impacts on salmonid populations in Washington (Scordino 2023). The WSAS concluded that the number of Chinook salmon eaten by seals and sea lions is “*substantial and has increased steadily over time*” (WSAS 2022). The WSAS also found that despite ongoing uncertainties in current understanding of salmon-pinniped interactions, given the large numbers of pinnipeds, even minimal predation can strongly impact low abundance and threatened salmon stocks. The “*preponderance of evidence supports the hypothesis that current populations of pinnipeds are likely impeding the recovery of salmon populations in Washington waters*” (WSAS 2022). The report also concluded that status quo pinniped management is inadequate to recover salmon and suggested strategic adaptive management approaches to improve salmon survival in the near term. For example, WSAS recommended targeted pinniped management at geographic locations with disproportionately high impacts on salmon populations, including the Ballard Locks, to support recovery of these highly impacted salmon populations.

Mike Mahovich, Muckleshoot Tribe Fisheries Manager, presented the current declining status of the salmon runs through the Ballard Locks and information on the more recent, increasing harbor seal predation on ESA-listed Chinook, sockeye, and coho salmon at the Ballard Locks. Numbers of sockeye have declined dramatically in the watershed, and Chinook populations remain far below recovery goals. Mike also described recent observations of harbor seals quickly learning to work around non-lethal deterrence efforts and forage effectively in the fish ladder, in the spillway, and in front of the large lock chambers.

Following the presentations, discussions at the workshop focused on what pinniped control measures worked and what did not at Bonneville Dam, Willamette Falls, and the 1980–1990s efforts at the Ballard Locks. The discussions drew heavily from management experiences and lessons learned over the past few decades at these constructed impoundments. The workshop addressed impoundment systems as a unique pinniped management context that differs from open waters. The focus of the workshop was on

measures to specifically address pinniped predation at the Ballard Locks, while acknowledging there are numerous vital forums and technical committees working on other limiting factors to salmon recovery in the Lake Washington watershed and throughout Puget Sound.

II. REVIEW OF PINNIPED CONTROL MEASURES AT REGIONAL IMPOUNDMENTS

The workshop reviewed the chronology of predation problems at the Ballard Locks, Bonneville Dam, and Willamette Falls and the effectiveness of measures to control predation on salmonids. The two sites in Oregon (Bonneville Dam and Willamette Falls) are currently the only sites nationally where the MMPA Section 120 authority to lethally remove pinnipeds has been implemented due to documented pinniped predation impacts on salmonids.

Joe Scordino (retired NOAA/NMFS), Kyle Tidwell (USACE), and Mike Brown (ODFW) provided an overview of the chronology of the pinniped issues and lessons learned at Ballard Locks (Scordino 2023), Bonneville Dam (Tidwell 2023), and Willamette Falls (Brown 2023), as summarized below. For more details on the deterrence methods discussed, see Appendix A.

A. Chronology of Pinniped Predation and Management Efforts at the Ballard Locks

The California sea lion and steelhead predation issue first came to the attention of the public in 1980 when a California sea lion (nicknamed “Herschel”) was observed preying on steelhead on a consistent basis in the spillway area of the Ballard Locks. Prior to 1980, sea lion presence at the Ballard Locks was rare, with the first known predation incident reported in 1970. California sea lions were not common in Puget Sound until the late 1970s, and as their occurrence in Washington State waters increased in the 1980s, so did their predation on steelhead at the Ballard Locks (NMFS and WDW 1989).

The occurrence of California sea lions preying on steelhead at the Ballard Locks was not viewed as a resource issue by fishery managers until 1985, when a serious decline in the Lake Washington winter-run steelhead population was noted. Extensive studies conducted by NOAA/NMFS and WDFW determined that predation by California sea lions at the Ballard Locks facility was a principal factor affecting the spawning escapement of returning adult winter-run steelhead in the Lake Washington basin (Jeffries and Scordino 1997, Pfeifer 1988, NMFS and WDW 1989). Although other factors such as freshwater and ocean survival likely contributed to steelhead declines, due to the small size of the total wild run (which historically averaged about 2,500 fish each year), sea lion predation was identified as a factor in the functional extirpation of the Lake Washington steelhead population (WDFW 2018).

The chronology presented below of the California sea lion and steelhead predation conflict (adapted from Scordino and Pfeifer 1993) illustrates the variety and intensity of management actions implemented in attempts to eliminate or minimize pinniped predation at the Ballard Locks.

This chronology includes more recent year observations from the workshop presentation by Mike Mahovich (Muckleshoot Indian Tribe). After the loss of the Lake Washington steelhead population, concerns about the pinniped conflict at the Ballard Locks shifted in the 2000s to the effects of increased harbor seal presence and predation on the declining Lake Washington sockeye and Chinook salmon populations. Although individual harbor seals at the Ballard Locks and in the Ship Canal have been observed sporadically year-round for many years, the occurrence has changed in recent years. Multiple harbor seals are now observed foraging successfully on salmon daily in and around the Ballard Locks and in the fish ladder (see the photo on the cover of this report). Most California sea lions are back at their Southern California breeding areas during the sockeye and Chinook salmon runs but do migrate back to Puget Sound in early fall when the coho salmon run is underway.

1980–1984. California sea lion predation on steelhead at the Ballard Locks was first noticed by the public in 1980. One large sea lion (“Herschel”) was observed on a daily basis at the entrance to the fish ladder in 1980–1981. State agents first started using firecrackers to harass sea lions out of the area in 1981. Acoustic harassment device and firecrackers were used by state agents in 1983/1984 to deter sea lions. One to three sea lions were observed in the spillway almost every week of the steelhead run in 1983/1984 (NMFS and WDW 1989).

1985–1986. The adverse effects of sea lion predation on steelhead escapement were first documented. Interagency monitoring and predation control program were initiated using chaser boats and underwater firecrackers. The number of sea lions foraging daily on steelhead at the Ballard Locks increased from four to six. WDFW begins predation monitoring and estimated that 15% of the steelhead run was consumed by sea lions (NMFS and WDW 1989).

1986–1987. Harassment using firecrackers continued, but the effectiveness in deterring sea lions declined drastically. Capture of sea lions in an entangling net was attempted unsuccessfully. Taste aversion conditioning using lithium chloride was attempted but ineffective. Intensified harassment efforts implemented late in the season involved long distance vessel chases and hazing, increased use of firecrackers, and use of an acoustic harassment device. Experimentation with the use of killer whale vocalizations was conducted but ineffective. The number of California sea lions foraging in the Ballard Locks spillway area increased to 8 to 10 individuals, the number of sea lions observed daily at the entrance to the Ship Canal in Shilshole Bay increased to 10 to 15 individuals, and some animals were reported in Lake Washington. Sea lion predation increased to 42% of the steelhead run (Gearin et al. 1986, 1988; Pfeifer 1987).

1987–1988. A barrier net was installed in the spillway near the fish ladder to prevent sea lion access to the main predation areas. The monitoring and harassment program was continued. Additional harassment techniques were tested using firecrackers in combination with boat hazing. The number of sea lions increased from 10 to 12 foraging daily at the Ballard Locks and 20 to 30 in Shilshole Bay. The barrier net did not reduce sea lion predation; rather, the predation shifted farther downstream. Several sea lions were observed preying on steelhead in Lake Washington; one animal remained above the dam through the summer and preyed on sockeye salmon as they exited the fish ladder. Sea lion predation increased to 52% of steelhead run, with steelhead escapement dropping to 858 fish (Pfeifer et al. 1989).

1988–1989. To obtain information on individual California sea lions and determine movement patterns and individual foraging behavior at the Ballard Locks, NOAA/NMFS and WDFW scientists captured and marked (using hot iron branding) sea lions on a trap at Shilshole Bay just outside the Ship Canal leading to the Ballard Locks. Thirty-nine of the branded sea lions were relocated to the outer coast of Washington (Long Beach peninsula). Over 75% returned to Puget Sound in an average of 15 days (ranging from 4 days to 45 days), and the effort therefore did not reduce predation. Ten to 12 sea lions foraged daily at the Ballard Locks through the steelhead run, and the number in Shilshole Bay increased to 30–40. Sport and tribal fisheries on steelhead in the Lake Washington system were closed to eliminate all take of wild winter-run steelhead. Sea lion predation exceeded 65% of the total run. Due to legal concerns about pinniped predation effects on tribal treaty rights, the U.S. District Court issued a stipulation that state, federal, and tribal parties must work together to address all aspects of the wild steelhead decline (Pfeifer 1989, NMFS et al. 1989).

1989–1990. Six sea lions were captured and relocated back to their breeding area off Southern California (Channel Islands). Attempts to capture more animals were unsuccessful because animals did not utilize the haul-out trap. Three of the six animals returned to Puget Sound—one in 30 days and the other two in approximately 45 days from release. A fourth animal returned as far as southern Washington (Columbia River). A tactile harassment program was attempted using rubber-tipped arrows to deter animals. An interagency technical committee on structural changes to the Ballard Locks facility was convened, but did not arrive at any structural modifications; the committee only made recommendations on fish passage studies and recommended modified water spill patterns over the dam. The recommended altered spill protocol was implemented. Sea lions consumed 59% of the steelhead run (Pfeifer 1991).

1990–1992. No predator control program was implemented as interagency emphasis shifted to fish enhancement efforts at the direction of a new, interim NOAA/NMFS Director in Washington D.C. Predation monitoring occurred only in the 1990/1991 season that estimated 59% of the steelhead run was consumed by sea lions. The amended spill protocol continued. Salinity of fishway attraction water and ambient salinity below the dam were monitored. Experimentation with illumination of the fishway to enhance nighttime fish passage was attempted, and salinity data collection commenced. Test results on illumination were inconclusive and confounded by technical problems and the low numbers of returning fish (Pfeifer 1991).

1992–1993. An acoustic barrier for keeping sea lions away from the fishway was tested intermittently. No consistent monitoring was conducted. Conclusions on the effectiveness of the acoustic devices were complicated by observations by AIRMAR (the manufacturer of the devices) that the devices may not have been operating at maximum efficiency because of algal and barnacle growth found on transducers. No predation monitoring occurred, and spawning escapement dropped to all-time low of 184 steelhead (Pfeifer 1994).

1993–1994. Predation monitoring was reinitiated at the request (and funding) from the Northwest Congressional delegation. A phased non-lethal deterrence and removal program was initiated with the use of an acoustic barrier, firecrackers, and capture and relocation. Three sea lions were relocated late in the season back to the Channel Islands. Spawning escapement dropped to all-time low of only 70 steelhead. Sea lions were also observed preying on downstream migrating salmonid smolts at the Ballard Locks (Pfeifer 1994).

1994–1995. The MMPA was amended in 1994 to consider the lethal removal of sea lions at the Ballard Locks. WDFW applied and received authorization under the amendments to lethally remove identifiable sea lions. In accordance with newly amended MMPA, a Pinniped Fishery Interaction Task Force was formed and recommended lethal removal of “predatory” sea lions under certain conditions. Predation monitoring and operation of an expanded acoustic barrier (eight omnidirectional devices on a cable and three unidirectional on the bottom) were underway all season. Underwater firecrackers were used on sea lions that entered the “ensonified” zone (NMFS and WDFW 1995).

In accordance with a Task Force recommendation, temporary captive holding during the steelhead run was undertaken with sea lion #17. Steelhead escapement was down to only 126 spawners and predation was only at 8% of the run, likely due to the reduced number of returning steelhead. During the peak of the smolt migration in mid-May 1995, sea lions foraged in the ensonified zone between 50% and 60% of the time that they were present in the Ballard Locks area. Sea lion #225, which killed the

most steelhead on any single day in 1994/1995, was captured on May 24, 1995 and, due to the lateness in the season, was not placed in captivity but rather transported to the Strait of Juan de Fuca and released. The same circumstances occurred with sea lion #87, the most frequently observed identifiable sea lion at the Ballard Locks in 1995, which was captured on June 17 and also released in the Strait of Juan de Fuca. Both sea lions #87 and #225 were subsequently observed at the Ballard Locks in June consuming downstream migrating smolts (NMFS and WDFW 1995).

Limited observations conducted during the fall coho salmon season in 1995 demonstrated that individual sea lions foraging extensively at the Ballard Locks during the steelhead run were also foraging at the Ballard Locks during the coho run. Resighting data on the sea lions branded to date indicated most sea lions did not remain long in the Shilshole Bay area, and only a small number actually entered the Ballard Locks area to forage; during the 1994/1995 season, only seven (3%) of the marked sea lions were sighted in the Ballard Locks area during the steelhead run (Foley 1996).

1995–1996. WDFW and NOAA/NMFS continued the use of an expanded acoustic barrier and firecrackers in the ensonified zone. Under a revised MMPA Section 120 Letter of Authorization, WDFW was authorized to remove predatory sea lions (defined as individually identified with brand marks that had previously been observed killing steelhead at the Ballard Locks). Three predatory sea lions were captured in May and identified for lethal removal. Vice President Gore intervened and had Sea World in Florida accept them for permanent captivity on May 23, 1996 (NMFS 1996; Scordino 1996, 1997).

1996–2001. The acoustic barrier was in place but not continually on (or functioning). Sea lion presence diminished as total steelhead escapement dropped to fewer than 50 winter-run steelhead in 2000 and did not recover (WDFW 2018). The cooperative WDFW-NOAA/NMFS pinniped predation monitoring program ceased (NMFS 2001).

2002–2015. With the loss of Lake Washington steelhead and subsequent reduced occurrence of California sea lions at the Ballard Locks, NOAA/NMFS involvement with pinniped control ceased. The MMPA Section 120 Letter of Authorization to WDFW expired in 2006 (NMFS 2002).

2016–2019. The Muckleshoot Indian Tribe observed increasing harbor seal presence and predation during the summer/fall salmon runs. The tribe requested assistance from NOAA/NMFS in 2017 and was advised that the prior year acoustic devices were no longer functional. No directed, continuous deterrence effort was conducted. Firecrackers were thrown opportunistically and sporadically when seals were in the fish ladder and appropriate co-manager staff were on-site. The lack of continuous deterrence efforts appeared to embolden harbor seal foraging behavior in the spillway area, allowing the seals to become “specialists” (i.e., habituated to foraging on salmon at the Ballard Locks).

2020–2023. Harbor seals were observed by Muckleshoot Indian Tribe biologists on a daily basis foraging on salmon in the fish ladder, in the spillway area, and downstream of the two lock chambers during the sockeye, Chinook, and coho salmon runs. Seal presence in the fish ladder increased significantly with up to seven individual seals observed in the ladder at one time. During sockeye salmon trapping operations, a number of salmon were observed with injuries associated with seal predation. These observations raised further concerns about the adverse effects of harbor seal foraging on salmon survival, including indirect mortality due to stress and injury from predator pursuits.

An acoustic device called TAST (Targeted Acoustic Startle Technology) was tested in the spillway area of the Ballard Locks during the Chinook and coho salmon runs in 2020. TAST proponents reported success

in reducing harbor seal predation in 2020, but it did not address increased seal predation inside the fish ladder nor any change in seal presence patterns. To enter the fish ladder, harbor seals had to swim in close proximity to the TAST device, thus indicating the device was not deterring foraging activity. Up to nine harbor seals were observed foraging at one time in the Ballard Locks area in 2021, and further TAST testing did not show the same success in reducing predation in 2021 as it did in 2020. The Muckleshoot Indian Tribe fish biologists reported increased numbers of harbor seals foraging in the fish ladder and resorted to using underwater firecrackers to deter the seals. The use of underwater firecrackers to keep seals out of the fish ladder became an issue in the analysis of TAST effectiveness (Williams et al. 2021, 2022). Field tests with the TAST in 2022 were aborted mid-season when firecrackers had to again be used to keep seals (undeterred by the TAST devices) out of the fish ladder.

2023. The Muckleshoot Indian Tribe and WDFW undertook a concerted deterrence effort using underwater firecrackers during the sockeye trap-and-haul operations to keep harbor seals out of the fish ladder and minimize predation in the spillway area. Seals reacted to the firecrackers by moving short distances or temporarily leaving the spillway area. Although the concerted deterrence effort appeared to have some effectiveness in keeping seals from entering the fish ladder, it was only effective in the short term while staff were actively throwing firecrackers. It did not eliminate harbor seal predation in the spillway area.

B. Chronology of Pinniped Predation and Management Efforts at Bonneville Dam

Bonneville Dam is located 146 miles upriver from the mouth of the Columbia River. Sporadic sightings of pinnipeds with a few instances of salmonid predation below the dam have been reported as early as the 1950s. In the early 1990s, concerns about pinniped predation on Columbia River salmonids increased when hatchery operators began reporting substantial increases in pinniped canine scarring on salmon (Scordino 1993), and open wounds caused by pinnipeds were observed on spring Chinook salmon in the Snake River (Harmon and Matthews 1990). During the 2001 migration of one of the largest runs of spring Chinook salmon to pass Bonneville Dam, the sightings of pinnipeds below the dam changed to daily sightings of several California sea lions capturing salmonids (Ward 2002).

2001. After several sea lions had been reported almost daily for a month in the Bonneville Dam tailrace, USACE began collecting data on sea lion presence. Observations began on April 11, 2001, and sea lions were observed over 70% of the days, until the last California sea lion was observed on May 13, 2001 (Stansell 2004).

2002. NMFS required USACE to monitor sea lions and their predation activities at Bonneville Dam. USACE began a concerted pinniped predation monitoring program and subsequently began working with ODFW, WDFW, CRITFC, and the U.S. Department of Agriculture (USDA) on a cooperative deterrence program (Stansell 2004).

2002–2004. USACE conducted daily pinniped monitoring during the spring run, with predation increasing from an estimated 1,010 adult salmonids (0.4% of the run) in 2002 to 3,533 salmonids (1.9% of the run) in 2004. Over 100 different individual sea lions were observed in 2003 and 2004, with a one-day maximum number in the tailrace exceeding 30 individual sea lions (Stansell 2004).

2005. To prevent sea lions from entering the fish ladders at Bonneville Dam, USACE designed removable “sea lion exclusion devices” (SLEDs) and installed them late in the season (about one week before sea

lions departed for the season) at four of the fish ladder entrances. The SLEDs consisted of vertical bars with 40-centimeter spacing to allow fish to pass but not sea lions. The USACE pinniped monitoring project estimated that 2,920 adult salmonids (3.4% of the run) were consumed by sea lions from January 1 to May 31, 2005 (Stansell 2005).

2006–2008. USACE continued the spring-run monitoring program, with estimated predation at 3.1%, 4.7%, and 3.2% in 2006, 2007, and 2008, respectively (Tidwell et al. 2022). The cooperative deterrence program was initiated in 2006, with USACE placing acoustic deterrence devices (ADDs) that emitted 205 dB at 10–15 kHz at the base of the fish ladder entrances during the spring runs each year. WDFW, ODFW, and CRITFC conducted boat-based deterrence each spring below the dam using above-water pyrotechnics (cracker shells and screamer/poppers), rubber batons or buckshot, and underwater firecrackers. In addition, pyrotechnic screamers and poppers, cracker shells, and rubber bullets/batons were used opportunistically from dam structures and land each spring by USDA wildlife agents to deter sea lions within 30 meters of fish ladder entrances. Sea lions were trapped at Bonneville Dam and relocated to Astoria in 2007, and all but one returned to Bonneville Dam within one to three weeks of release. Six sea lions were captured and transferred to Sea World for permanent captivity (Wright et al. 2007; Tackley et al. 2008; NMFS 2008; Brown et al. 2007, 2008).

The SLEDs were installed at all of the main fish ladder entrances by early February in 2006 and beyond. Only one sea lion, branded #C404, was observed in the fish ladder in 2006 and 2007, while all other sea lions appeared to have been excluded by the SLEDs (Stansell 2006).

On November 11, 2006, ODFW, WDFW, and the Idaho Department of Fish and Game (IDFG) applied for MMPA Section 120 authorization to remove California sea lions at Bonneville Dam (ODFW et al. 2006). The states' request for this authorization was based on observations of individual branded California sea lions, indicating that many of the same sea lions were returning year after year (the inter-year return rate was greater than 70%), and some were spending more time each year foraging at Bonneville Dam. Steller sea lions (*Eumetopias jubatus*), which were relatively easy to deter in early years, became bolder and less responsive to deterrence (ODFW et al. 2006). But it was not until March 18, 2008, that NOAA/NMFS issued a 5-year authorization under MMPA Section 120 for the permanent removal of individually identifiable California sea lions that repeatedly foraged on salmonids at Bonneville Dam (NMFS 2008). Six California sea lions were captured at Bonneville Dam and transferred to permanent captivity at Sea World under this new MMPA Section 120 authority.

The 2008 ESA Biological Opinion on the USACE operation of the federal dams (i.e., Bonneville Dam) required USACE to monitor sea lion abundance and predation attempts/rates, and to assess the effectiveness of deterrence methods which, in combination with other environmental and operational data collected by USACE and cooperating agencies, provided a good example of comprehensive monitoring necessary for adaptive pinniped management (NMFS 2008).

2009. The USACE monitoring program documented an estimated predation of 4,489 adult salmonids (2.4% of the run). Sea lions were observed swimming and eating salmon within 20 feet of the ADDs. Pinniped hazing from boats and the dam continued. A total of 21 California sea lions were captured with 11 euthanized, four moved to permanent captivity, and six released in accordance with Section 120 authorization (Stansell et al. 2010, Brown et al. 2009).

While the SLEDs kept sea lions out of the fishways, other non-lethal deterrents had only a temporary effect. Boat-based hazers from CRITFC used a combination of deterrents (e.g., firecrackers, cracker shells, and vessel chase), and USDA continued hazing from dam structures. The purpose of non-lethal deterrent activities was to: (1) attempt to disrupt sea lion foraging behavior and reduce sea lion abundance immediately below Bonneville Dam, (2) attempt to discourage naïve animals from becoming habituated to foraging below the dam, and (3) to satisfy the Section 120 requirements for hazing prior to any removal efforts. However, based on field observations through the years, there was no obvious reduction in overall sea lion abundance or predation in response to hazing (Brown et al. 2009).

2010–2019. California sea lions were sporadically removed over the following years, as this authorization was repeatedly challenged in court. In 2013, the appeals court ruled in favor of NOAA/NMFS, and permanent removal of sea lions continued (Edwards et al. 2022). This authorization was extended several times through 2021, allowing ODFW and WDFW to trap and euthanize California sea lions at the dam that met certain criteria. The authorization did not apply to Steller sea lions, however, whose occurrence and residency were increasing each year along with a change in behavior to increasingly bold and oblivious to deterrence.

ODFW and WDFW permanently removed 221 California sea lions. The permanent removals did reduce the presence of and salmonid predation by California sea lions. However, the Section 120 authorization did not apply to Steller sea lions, and by 2017 their salmonid consumption exceeded that of California sea lions. By 2019, Steller sea lions were the most abundant salmonid predator at Bonneville Dam. Up until 2017, the monitoring program focused on the spring salmon runs at Bonneville Dam. However, the predation monitoring was expanded due to increasing observations of Steller sea lion predation during the fall/winter (Stansell et al. 2011, 2012, 2013, 2014; Brown et al. 2010, 2011, 2012, 2013; USACE 2015; NMFS 2016; Tidwell et al. 2017, 2019, 2021; Madson et al. 2017; Steingass et al. 2019).

2020–2022. As a result of the 2018 Amendments to the MMPA, a new MMPA Section 120(f) authorization was issued on August 14, 2020 that provided for lethal removal of both California and Steller sea lions in the Columbia River and its tributaries without the restrictive criteria of identifying individual animals that hampered the effectiveness of the prior permanent removal approach. The 2020 Federal Columbia River Power System Biological Opinion (NMFS 2020) under the ESA continued the requirement for USACE to monitor the seasonal presence, abundance, and predation activities of sea lions at Bonneville Dam. With the new authorization, the states removed 55 Steller sea lions and 35 California sea lions (NMFS 2020; Clark et al. 2021; Brown et al. 2021, 2022; Edwards et al. 2022; Tidwell and van der Leeuw 2021; Steingass et al. 2020; van der Leeuw and Tidwell 2022).

2023. Permanent removals continued, with six Steller sea lions and 22 California sea lions removed through the spring.

C. Chronology of Pinniped Predation and Management Efforts at Willamette Falls

Willamette Falls is located 128 miles upriver from the mouth of the Columbia River at Oregon City, Oregon. The earliest known report of California sea lions at this site was in the 1950s (Beach et al. 1985), and then in April 1975, when two sea lions were reported taking salmon and hindering fish passage at the fish ladder (NMFS and ODFW 1997). Other than the 1975 sighting, there were no reports of sea lions at Willamette Falls until the late 1980s when personnel at the fish ladder reported California sea lion sightings below the falls. Sea lions were sighted sporadically near the falls until 1995, when California sea

lions began occurring almost daily from February through late May. The year 1995 also marked the first year that a California sea lion was reported repeatedly entering one of the fish ladders and consuming spring Chinook salmon (NMFS and ODFW 1997).

1995. ODFW began a preliminary pinniped predation monitoring program after receiving repeated reports of a California sea lion preying on salmonids near and in the fish ladders at Willamette Falls. ODFW observed four different California sea lions (identified by unique markings and size) foraging in the area until May 24, when the last sea lion was observed. Fifteen predation events were observed (nine spring Chinook, two winter steelhead, and four unidentified salmonids) (NMFS and ODFW 1997).

1996. ODFW and NOAA/NMFS developed a joint monitoring plan, and NOAA/NMFS provided funding to ODFW to initiate a daily monitoring program during the spring Chinook salmon and winter steelhead migration up the Willamette River. ODFW conducted 210 hours of daylight surface observation from April 2 to May 31, 1996. Seven different California sea lions were observed on 26 of the 46 days of observation, with up to four sea lions observed in the area at the same time. ODFW observed 115 predation events by at least seven different California sea lions consuming 89 salmonids (Scordino 2010).

ODFW installed a barrier gate at fish ladder entrance #1, where a sea lion had entered the fish ladder in 1995. The barrier was constructed of 2.5-centimeter aluminum bar stock, with 3-meter vertical bars spaced at 38-centimeter intervals placed at the bottom of the fish ladder with a stop-log gate lowered on top of it. The removable barrier was designed to allow unrestricted access by adult salmonids and prevent access by sea lions. Observations by ODFW in 1996 indicated that the barrier effectively prevented sea lions from entering the fish ladder (Brown et al. 1998).

1997–1999. ODFW continued the spring monitoring program and used firecrackers and rubber projectiles shot from a shotgun to deter sea lions from areas near the fish ladder entrances. These deterrents caused sea lions to temporarily move away from the immediate area and changed their pattern of approach, but the deterrence did not prevent continued foraging. A floating baited trap was placed near one of the fish ladder entrances in an attempt to capture the problematic California sea lions, but they did not enter the trap (Boatner 2000). The estimated salmonid predation was 193, 215, and 356 in 1997, 1998, and 1999, respectively; and the percentage of the runs taken by sea lions was 0.5%, 0.2%, and 0.4% of the spring Chinook run and 1.5%, 3.5%, and 3% of the winter steelhead run (Boatner 1997, Brown et al. 2004).

2000–2002. ODFW continued the monitoring program and deterrence efforts (with limited effectiveness). Paint balls dispatched from a CO₂ pistol were used in 2000 to deter and temporarily mark sea lions. The effects were similar to the other deterrents in that the struck sea lions would move from the immediate area but continued foraging. One branded sea lion (also seen in 1999) was frequently observed eating salmon. The estimated total predation in 2002 was 185 salmonids, with an estimated 0.2% of the spring Chinook run and 0.3% of the winter steelhead run (Brown et al. 2004).

ODFW tested the use of rubber projectiles (rubber buckshot and batons) shot from a shotgun to deter the California sea lions observed to repeatedly forage near the fish ladders. Sea lions were identified by natural markings or by temporary streamer dart tags that were applied using a CO₂ pistol. Four individually recognizable sea lions were shot with rubber buckshot and batons. The smaller and less commonly occurring sea lions reacted more to being shot than the larger, bolder sea lions. The smaller

animals immediately left the area and did not return, while the larger animals returned to the immediate area in less than 24 hours. On return, the larger sea lions were more wary, moving from area to area, surfacing in less predictable ways, and spending more time underwater, thus preventing dispatch of another rubber projectile (Boatner 2000).

2003. ODFW ceased its monitoring program and utilized an all-volunteer observation program to collect more qualitative data. Up to four sea lions were observed in the area at the same time. The estimated percent of the runs taken by sea lions in 2003 was 0.1% of the spring Chinook run and about 1% of the winter steelhead run (Brown et al. 2004).

2004–2009. ODFW did not conduct monitoring or hazing because it determined that the overall impact of sea lion predation at this site since 1996 had been low and may not be of great concern for salmon passage or survival. However, Portland State University students conducted some monitoring, which indicated that an increase in predation activity by California sea lions was occurring below the falls (Wright et al. 2014).

2010–2013. Oregon State Legislature passed a bill in June of 2011 requiring ODFW to conduct hazing of sea lions (OR House Bill HB3255). Vessel chases and hazing occurred in 2010, 2011, and 2013. While hazing was effective at moving California sea lions downstream away from the fish ladder entrances, once hazing stopped, predator activity would resume. It is unknown whether hazing had any significant effect on the total predation rate of winter steelhead (Wright et al. 2014).

2014–2017. Monitoring was conducted but no hazing. Annual California sea lion numbers peaked in 2017 at over 40 sea lions on a single day (Wright et al. 2017). The Upper Willamette River Winter Steelhead Population Viability Study conducted by ODFW scientists in 2017 concluded that the upper Willamette River native (winter) steelhead were at significant risk of extinction due to predation by California sea lions (Falcu 2017). An estimated 25% of the potential escapement of winter steelhead was consumed by California sea lions (Wright et al. 2014, 2015, 2016, 2017). ODFW determined that the problem of California sea lions taking ESA-listed salmonids below Willamette Falls was significant, and the state applied for an MMPA Section 120 authorization on October 5, 2017 (ODFW 2017, NMFS 2018).

2017–2018. ODFW conducted a capture and relocation effort that only temporarily removed the sea lions (they quickly returned). ODFW received an MMPA Section 120 authorization for lethal removal of California sea lions but not until November 14, 2018 (NMFS 2018). Fishway SLEDs were replaced in 2017 due to the original grates becoming damaged and/or being overtopped at high water levels, allowing sea lions to once again enter the fish ladders (NMFS 2018, Wright et al. 2017).

2018–2019. ODFW permanently removed 33 adult California sea lions between December 2018 and May 2019 under MMPA Section 120 authority (Steingass et al. 2019, Wright et al. 2018).

2019–2020. A new MMPA Section 120(f) authorization provided for removal of both Steller and California sea lions. This new authorization issued 8/14/20 also removed restrictive criteria for the removal of individual sea lions. Due to COVID, there was no activity in early 2020, but six Steller sea lions were lethally removed in October and November 2020 when activities resumed (Steingass et al. 2020, Wright et al. 2020).

2020–2021. ODFW lethally removed eight California sea lions [six under new 120(f) authority and two under authority granted in 2018] and one Steller sea lion [taken under new 120(f) authority] (Brown et

al. 2022). ODFW reported substantial decreases in predator abundance and associated salmonid predation compared to pre-management years (prior to 2018 lethal removal authorization), particularly for ESA-listed winter steelhead (Wright et al. 2021).

2022. ODFW lethally removed three California sea lions (one-half of the estimated sea lions observed). Removals have become increasingly difficult due to a decrease in trap use by sea lions (Edwards et al. 2022, Wright et al. 2022).

2023. ODFW reported that at least six California sea lions and two Steller sea lions were foraging at Willamette Falls in early 2023 and could not be removed due to lack of trap use. [Note: the MMPA Section 120(f) authorization does not allow shooting sea lions in the water.]

III. RECOMMENDATIONS FOR PINNIPED MANAGEMENT AT THE BALLARD LOCKS

Management recommendations developed during the workshop are organized into four categories, which should be implemented concurrently to address pinniped predation on salmon at the Ballard Locks. Effective management interventions must include a multi-pronged adaptive approach of pinniped removal, deterrence, and exclusion as well as operational adjustments to enhance salmon migration. The immediate removal of specialist predators from a target area should co-occur with a deterrence effort to prevent naïve pinnipeds from learning specialist predator behaviors at the site. A comprehensive, real-time monitoring component is essential for an adaptive approach to addressing pinniped predation.

- **Pinniped Removal** – Permanently remove the pinnipeds that occur in the area from the Salmon Bay Railroad Bridge to the Ballard Locks (as shown in Appendix B) that are not responsive to deterrence efforts.
- **Pinniped Deterrence** – Implement a multifaceted deterrence program during upstream and downstream salmon migrations to prevent pinnipeds from habituating to foraging at the Ballard Locks (i.e., deter newly arriving or naïve pinnipeds so that they do not become specialists that must be permanently removed).
- **Structural and Operational Changes at Facility** – Design and install a pinniped exclusion structure at the entrance to the fish ladder that prevents pinniped access without affecting salmon passage (similar to the SLEDs/barrier gates installed at Bonneville Dam and Willamette Falls). Continue studies on the feasibility of structural and operational modifications to enhance salmon migration through the Ballard Locks.
- **Comprehensive Monitoring** – Implement a comprehensive monitoring program focused on salmon passage; pinniped presence, behavior, and predation; and environmental and operational variables at the Ballard Locks. The data collection regimes should be structured to inform, on a real-time basis, the implementation of adaptive management measures to minimize or eliminate the pinniped-salmonid conflict at this site. Monitoring should occur during both adult and smolt migration periods for sockeye, Chinook, and coho salmon.

Additional actions recommended to enable meaningful progress on the above management actions are as follows:

- **Outreach and Education** – Implement a public relations campaign focused on the detrimental effects of pinniped predation on salmon at the Ballard Locks and the urgent need for effective measures to control pinniped predation on salmonids.
- **ESA Consultation on Ballard Locks** – Assess the effects of pinniped foraging on salmon passage survival and evaluate additional physical and operational measures to control it in the next Biological Assessment necessary for ESA consultation on the effects of the Ballard Locks facility and operations on ESA-listed Chinook salmon.

Lessons learned and recommendations are described for each component of the four-pronged management approach to pinniped management at the Ballard Locks. For more details on the deterrence methods, see Appendix A.

A. Lessons Learned and Recommendations – Pinniped Removal

- Some individual pinnipeds will exhibit specialist feeding behaviors to effectively prey primarily on salmon. These individuals – called “specialists” – can have the greatest impacts on local salmon populations. Removal of problematic individual specialist seals and sea lions is essential to long-term salmon recovery objectives.
- Relatively few of the sea lion specialists were responsible for much of the pinniped predation on salmonids at constructed impoundments, including the Ballard Locks. This is likely true for the harbor seal specialists also.
- Removal is only proposed for individual seals and sea lions foraging within the confined area upstream of the Salmon Bay Railroad Bridge to the Ballard Locks (see Appendix B).
- If other activities in the vicinity are attracting pinnipeds, those activities must be addressed to reduce new recruits to the impoundment predation conflict.
- General removals, removals of new or naïve animals, or removals of animals outside the target vicinity will not be effective.
- Pinnipeds appear to transmit or “teach” specialist feeding behaviors to one another. Ideally, removal of specialists would be the first step in the management sequence, swiftly followed by and combined with deterrence to prevent new, naïve individuals from learning to feed on salmon at the Ballard Locks, as well as structural and operational changes that improve fish passage and/or exclude pinnipeds from fish ladders. Swift timing is important to prevent new naïve pinnipeds from becoming adapted to the site.
- Removal is challenging, both from a public education/relations perspective, as well as from a regulatory perspective (in particular, the MMPA currently imposes cumbersome processes and constraints on lethal removals).
- A robust public education and relations program is needed to foster public support and reduce public harassment of co-managers and other staff working on-site at the Ballard Locks. Public access to the Ballard Locks will need to be controlled during any actual removal efforts.
- Removal methods include humane euthanasia and permanent transfer of individuals to aquariums and zoos for captive display.
- Relocating problematic individuals (as an alternative to lethal removal) has proven ineffective. Sea lions that were relocated several hundred miles away from impoundments quickly returned to the impoundment. One harbor seal captured at the Ballard Locks and relocated to Hood Canal in 1993/1994 showed similar behavior in returning to the Ballard Locks.
- Requirements to provide definitive evidence that individual pinnipeds are having a significant impact prior to removal will cause management delays that result in further declines or losses of at-risk salmon stocks.
- Knowledge and experience with pinniped conflicts at previous and ongoing pinniped management interventions should be applied so that managers do not have to “recreate the wheel” whenever a pinniped conflict is noted.

B. Lessons Learned and Recommendations – Pinniped Deterrence

- Although deterrence actions alone have limited effectiveness, it is important to continue to implement and incorporate them into multi-pronged management approaches moving forward.
- Pinnipeds new to an impoundment (naïve pinnipeds) are thought to be more susceptible to non-lethal deterrence methods than animals that have already habituated to foraging on salmon at a specific location. Once pinnipeds have learned the benefits of foraging at impoundments (i.e., easy access to prey), deterrence techniques are less effective.

- Proactive approaches to discourage new animals from entering an area and learning its benefits (i.e., easy availability of prey species) are critical to preclude naïve animals from becoming specialists.
- Deterrence alone may not prevent habituation. Some pinnipeds will become specialist feeders on salmon at certain sites regardless of deterrence measures.
- There is a need to consider the behavioral cues and training signals of individual pinnipeds – it is not possible to influence or train a population.
- Primary deterrence techniques include noise stimuli (see Appendix A) that have been shown to initially cause pinnipeds to avoid or flee from an area.
- In the 1980s and 1990s, harbor seals were easily spooked by deterrence measures compared to the sea lions. Years later, however, the harbor seals now foraging at the Ballard Locks are exhibiting the same avoidance and immediate return behaviors initially only observed with sea lions in the past.
- Public outreach and education materials, including signage at the Ballard Locks, should contain information about the impacts of pinniped predation on ESA-listed and at-risk salmon species.
- Use of underwater firecrackers tends to move pinnipeds away from the immediate area and only for a short time, but the animals do not leave the Ballard Locks vicinity.
- Pinnipeds can be hazed from the shore or from a boat, although accurate hazing from shore can be a challenge. The animals also learn to avoid individuals they associate with hazing (e.g., someone wearing an orange jacket).
- To date, acoustic deterrence methods have been mostly ineffective; the specialist pinnipeds animals quickly learn to tolerate the noise or swim with their heads above the water, avoiding the underwater noise.
- Seals and sea lions learn and adapt. The workshop recommended using engagement and advice of animal behavior experts. In conjunction with removal, varied and novel deterrence may reset the dynamic and potentially deter naïve animals from learning specialist feeding behavior.
- There is a need to carefully monitor potential effects of deterrence actions on salmon behavior, especially when using noise stimuli that might also affect salmon behavior.

C. Lessons Learned and Recommendations – Structural and Operational Changes at the Ballard Locks

- Barriers to fish passage and migration delays increase salmon exposure and vulnerability to predation and injury by pinnipeds.
- The existing fish ladder is a migration bottleneck at the Ballard Locks. Seals and sea lions gather near the downstream entrance to the ladder and predate on adult migrating salmon, which tend to pause and hold at the locks and ladder entrance. Some seals will swim into the fish ladder itself, where salmon are completely trapped and vulnerable to easy predation, injury, or stress.
- The fish ladder at the Ballard Locks was initially installed in 1917, and it was upgraded/replaced in 1976 to improve passage conditions. Since that time, there have been advancements in fish ladder design. The workshop recommends further consideration of ladder redesign for improved fish passage and prevention of pinniped foraging.
- At Bonneville Dam and Willamette Falls, sea lion exclusion devices (SLEDs/barrier gates) were designed and installed at the fishway entrances to prevent pinnipeds from entering the fishways; thus, directly preventing pinniped predation on salmonids in the fish ladders. The SLED design is a vertical metal plate or frame at the entrance to the ladder, with metal bars spaced so salmon can enter but seals and sea lions cannot. The workshop strongly recommends designing and installing a similar device to exclude harbor seals at the Ballard Locks fish ladder as an

immediate approach to eliminate salmon stress and predation by pinnipeds within the ladder. Key design elements of the fish ladder exclusion device include bar spacing and shape (e.g., round vs. flat bars), hydraulic effects (such as vibration), installation methods and constraints, bar spacing to target different species' sizes (salmon, seals, sea lions), maintenance needs, and debris trapping. Ease of removal is critical; the exclusion device should be able to be removed immediately and modified relatively quickly if monitoring suggests its design is problematic for migrating salmon.

D. Lessons Learned and Recommendations – Comprehensive Monitoring

- An adaptive management approach with robust real-time monitoring is essential. Pinniped and salmon behaviors may change in response to predation management measures and changing site conditions (tides, spill, locks/spillway operations, structure changes, etc.).
- Comprehensive monitoring is also essential in determining if pinniped deterrence appears successful but is really an artifact of prey presence or availability, environmental conditions (e.g., tidal movements), or other factors that may mask the interpretation of the deterrence data. Further, such monitoring data should address if the deterrence effort just shifts the predation location or causes other unforeseen circumstances.
- Pinniped predation affects both adult and juvenile salmon life stages. Focusing on only one pinniped interaction can result in ineffective long-term management and other unintended consequences. Methods that reduce predation on adult salmon likely benefit juvenile salmon too, but that should be monitored to validate.
- Harbor seals and sea lions may differ in predation behaviors and characteristics – there is a need to monitor and report real-time data in order to understand and appropriately adapt management interventions to effectively address both species.
- Any operational adjustment or management intervention should be monitored for potential effects on salmon migration delays and predation escape behaviors.

IV. REFERENCES AND ADDITIONAL SOURCE MATERIAL

As part of the workshop, a reference library on pinniped predation and deterrence methods was developed and made accessible to workshop participants. An index to the library, with links to complete copies of each document, can be accessed through request to WRIA 8 staff.

The reference library documents cited in this Recommendations Report are as follows.

- Beach, R.J., A.C. Geiger, S.J. Jeffries, S.D. Treacy and B.L. Troutman. 1985. Marine mammals and their interactions with fisheries of the Columbia River and adjacent waters, 1980-1982. NWAFC Process. Rep. 85-04. 316pp.
- Boatner, R. 1997. 1997 Willamette Falls sea lion observations. ODFW Progress Report to NMFS-NWR.
- Boatner, R. 2000. Willamette Falls sea lion observation project. ODFW Progress Report to NMFS-NWR.
- Brown, M. 2023. Lessons learned from Willamette and Bonneville Pinniped Management – ODFW perspective. Presentation at Day 2 of the Technical Workshop Series: Management of Pinniped Predation on Salmon at the Ballard Locks.
- Brown, M., S. Steingass, B. Wright, S. Riener, S. Valentine, Z. Kroneberger, B. Triplett, B. Phibbs, J. Burco, and C. Gillin. 2021. Willamette Falls Management Report ODFW Report.
- Brown, M., B. Wright, S. Valentine, S. Riener, Z. Kroneberger, E. Nass, J. Burco, and C. Gillian. 2022. Annual Report: Pinniped Management at Willamette Falls, 2021-2022. ODFW Report.
- Brown, R.F., D. Bennett, and R. Boatner. 1998. Monitoring California sea lion predation at Willamette Falls, Oregon, 1995-96. ODFW Tech. Rep. No. 96-6-0_ (Draft).
- Brown, R.F., B.E. Wright, and J. Hoehne. 2004. Monitoring predation by California sea lions and salmonid passage at Willamette Falls. Progress report 7/01/02-12/31/03. ODFW Progress Report to PSMFC.
- Brown, R.F., S.J. Jeffries, B.E. Wright, M. Tennis, P.J. Gearin, S.D. Riener, and D. Hatch. 2007. Field report - 2007 pinniped research and management activities at Bonneville Dam. ODFW Report.
- Brown, R.F., S.J. Jeffries, D. Hatch, and B.E. Wright. 2008. Field report: 2008 pinniped management activities at Bonneville Dam. ODFW Report.
- Brown, R.F., S.J. Jeffries, D. Hatch, B.E. Wright, S. Jonker, and J. Whiteaker. 2009. Field report: 2009 pinniped management activities at and below Bonneville Dam. ODFW Report.
- Brown, R.F., S.J. Jeffries, D. Hatch, B.E. Wright, and S. Jonker. 2010. Field report: 2010 pinniped management activities at and below Bonneville Dam. ODFW Report.
- Brown, R.F., S.J. Jeffries, D. Hatch, B.E. Wright, and S. Jonker. 2011. Field report: 2011 pinniped research and management activities at and below Bonneville Dam. ODFW Report.
- Brown, R.F., S.J. Jeffries, D. Hatch, B.E. Wright, and S. Jonker. 2012. Field report: 2012 pinniped research and management activities at and below Bonneville Dam. ODFW Report.

- Brown, R.F., S.J. Jeffries, D. Hatch, B.E. Wright, and S. Jonker. 2013. Field report: 2013 pinniped research and management activities at Bonneville Dam. ODFW Report.
- Carretta, J.V., E.M. Oleson, K.A. Forney, D.W. Weller, A.R. Lang, J. Baker, A.J. Orr, B. Hanson, J. Barlow, J.E. Moore, M. Wallen, and R.L. Brownell. 2023. U.S. Pacific marine mammal stock assessments: 2022. U.S. Department of Commerce, NOAA Technical Memorandum NMFS-SWFSC-684.
- Clark, C., M. Brown, D. Hatch, and J. Dupont. 2021. Final Field Report: 2017-2021 Pinniped Research and Management Activities at Bonneville Dam. Report to NOAA/NMFS.
- Cram, J., N. Kendall, A. Marshall, T. Buehrens, T. Seamons, B. Leland, K. Ryding and E. Neatherlin 2018. Steelhead At Risk Report: Assessment of Washington's Steelhead Populations. Washington Department of Fish and Wildlife Report FPT 19-03.
- Edwards, J., B. Wright, C. Clark, M. Brown, S. Valentine, D. Hatch, and J. Powell. 2022. Annual Report: 2022 Columbia River Basin Research and Management Activities. WDFW Letter to NOAA/NMFS on Section 120 Authorization.
- Falcy, M. 2017. Population viability of Willamette River winter steelhead - An assessment of the effects of sea lions at Willamette Falls. ODFW Report.
- Foley, S. 1996. Monitoring California sea lion predation in the Lake Washington Estuary in 1994-95. WDFW Report.
- Gearin, P.J., R. Pfeifer, and S.J. Jeffries. 1986. Control of California sea lion predation of winter-run steelhead at the Hiram M. Chittenden Locks, Seattle, December 1985-April 1986. WDG Fish. Mgmt. Report #86-20. 108pp.
- Gearin, P.J., R. Pfeifer, S.J. Jeffries, R.L. DeLong, and M. Johnson. 1988. Results of the 1986-1987 California sea lion-steelhead trout predation control program at the Hiram M. Chittenden Locks. NWAFC Process. Rep. 88-30. 111pp.
- Harmon, J.R., and G. Matthews. 1990. Evidence of increase in marine mammal damage to adult spring Chinook salmon in Columbia River. NWAFC Quart. Rep. April-May-June 1990.
- Jeffries, S.J., and J. Scordino. 1997. Efforts to protect a winter steelhead run from California sea lion predation at the Ballard Locks. In: Stone, G., J. Goebel and S. Webster (eds). Pinniped Populations, Eastern North Pacific: Status, Trends, and Issues. Symposium of the 127th Annual Meeting of AFS.
- Madson, P.L., B.K. van der Leeuw, K.M. Gibbons, and T.H. Van Hevelingen. 2017. Evaluation of pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2016. USACE Report.
- NMFS (National Marine Fisheries Service). 1996. Environmental assessment on conditions for lethal removal of California sea lions at the Ballard Locks to protect winter steelhead. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries Service). 1999. Report to Congress: Impacts of California sea lions and Pacific harbor seals on salmonids and west coast ecosystems. U.S. Dept. Commerce Report to Congress.

- NMFS (National Marine Fisheries Service). 2001. Environmental assessment on extending the letter of authorization for lethal removal of California sea lions at the Ballard Locks to protect Lake Washington winter steelhead. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries Service). 2002. NMFS decision memo on Extending LOA NMFS internal memorandum (January 22, 2002).
- NMFS (National Marine Fisheries Service). 2008. MMPA Section 120 Letter of Authorization. NMFS Letter of Authorization to ODFW, WDFW and IDFG.
- NMFS (National Marine Fisheries Service). 2008. Final environmental assessment: reducing the impact on at-risk salmon and steelhead by California sea lions in the area downstream of Bonneville Dam on the Columbia River, Oregon and Washington. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries service). 2008. Endangered Species Act Section 7(a)(2) Consultation Biological Opinion and Magnuson-Stevens Fishery Conservation and Management Act Essential Fish Habitat Consultation. Consultation on Remand for Operation of the Federal Columbia River Power System, 11 Bureau of Reclamation Projects in the Columbia Basin and ESA Section 10(a)(1)(A) Permit for Juvenile Fish Transportation Program (Revised and reissued pursuant to court order, NWF v. NMFS, Civ. No. CV 01-640-RE (D. Oregon)). NMFS Biological Opinion.
- NMFS (National Marine Fisheries Service). 2016. Supplemental Environmental Assessment: Reducing the Impact on At-risk Salmon and Steelhead by California Sea Lions in the Area Downstream of Bonneville Dam on the Columbia River, Oregon and Washington. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries Service). 2018. MMPA Section 120 Letter of Authorization. NMFS letter to ODFW.
- NMFS (National Marine Fisheries Service). 2018. Environmental Assessment. Reducing the Impact on At-Risk Salmon and Steelhead by California Sea Lions in the Willamette River. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries Service). 2020. Environmental Assessment. Reducing Predation Impacts on At-Risk Fish by California and Steller Sea Lions in the Columbia River Basin. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries Service). 2020. Endangered Species Act Section 7(a)(2) Biological Opinion and Magnuson-Stevens Fishery Conservation and Management Act Essential Fish Habitat Response for the Continued Operation and Maintenance of the Columbia River System. NMFS Biological Opinion.
- NMFS (National Marine Fisheries Service) and ODFW (Oregon Department of Fish and Wildlife). 1997. Environmental assessment of preventing California sea lion foraging and predation on salmonids at Willamette Falls, Oregon. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries Service) and WDFW (Washington Department of Fish and Wildlife). 1995. Environmental assessment on protecting winter-run wild steelhead from predation by

- California sea lions in the Lake Washington Ship Canal, Seattle, Washington. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries Service) and WDW (Washington Department of Wildlife). 1989. Environmental assessment on controlling California sea lion predation on wild steelhead in the Lake Washington Ship Canal. NMFS Environmental Assessment.
- NMFS (National Marine Fisheries Service), WDW (Washington Department of Wildlife), USACE (U.S. Army Corps of Engineers), Muckleshoot Tribe, and Suquamish Tribe. 1989. Interagency plan to conserve the 1988/89 winter run of wild steelhead into the Lake Washington drainage. NMFS and WDW Report.
- ODFW (Oregon Department of Fish and Wildlife). 2017. Request for Marine Mammal Protection Act Section 120 Authorization to Remove California Sea Lions from the Willamette River. Section 120h Application to NMFS (October 5, 2017).
- ODFW (Oregon Department of Fish and Wildlife), WDFW (Washington Department of Fish and Wildlife), and IDFG (Idaho Department of Fish and Game). 2006. Request for MMPA Section 120 Pinniped Removal Authority – Application. 11/13/06 Section 120h Application to NMFS.
- Pamplin, N., K. Lee, S. Pearson, and J. Anderson. 2021. Pinniped Conservation and Management. WDFW PowerPoint presentation.
- Pfeifer, R. 1987. Managing around marine mammal predation on winter-run steelhead returning to Lake Washington. WDW Fish. Mgmt. Div. #87-7.
- Pfeifer, R. 1988. Lake Washington winter-run steelhead/marine mammal interactions, problem history, past procedure used, and a discussion of possible control options. U.S. District Court-Ordered Interagency Task Force Report. WDW. 54 p.
- Pfeifer, R. 1989. Monitoring of the 1988-89 California sea lion control program in the Lake Washington Estuary. WDW Fish. Mgmt. Report #89-12.
- Pfeifer, R. 1991. Monitoring of 1990-91 California sea lion predation in the Lake Washington estuary. WDW Fish. Mgmt. Report #91-19.
- Pfeifer, R. 1991. Monitoring of the 1989-90 California sea lion control program in the Lake Washington Estuary. WDW Fish. Mgmt. Report #90-17.
- Pfeifer, R. 1994. Evaluation of acoustic deterrent devices and monitoring of 1992-93 steelhead predation by California sea lions in the Lake Washington estuary. WDFW Report.
- Pfeifer, R. 1994. Monitoring of 1993-94 California sea lion predation in the Lake Washington estuary. WDW Report.
- Pfeifer, R., P.J. Gearin, S.J. Jeffries, M. Johnson, and R.L. DeLong. 1989. Evaluation of the 1987-88 California sea lion control program in the Lake Washington Estuary. WDW Fish. Mgmt. Report #89-9.
- Scordino, J. 1993. Review and evaluation of pinniped predation on salmonids in the Columbia River Basin. NOAA/NMFS Northwest Region Report.

- Scordino, J. 1996. Factual Statement on California Sea Lion (“Herschel”) Conflict with Wild Steelhead at the Ballard Locks in Seattle, WA. NOAA/NMFS Factual Statement submitted to U.S. District for HSUS v. Dept. of Commerce.
- Scordino, J. 1997. Preliminary summary of 1997 observations on the sea lion-steelhead conflict at the Ballard Locks. NMFS-NWR Report.
- Scordino, J. 2010. West coast pinniped program investigations on California sea lion and Pacific harbor seal impacts on salmonids and other fishery resources. PSMFC Report.
- Scordino, J. 2023. Lessons learned from pinniped management at the Ballard Locks Presentation at Day 2 of the Technical Workshop Series: Management of Pinniped Predation on Salmon at the Ballard Locks.
- Scordino, J. 2023. Overview of pinniped management at the Ballard Locks Presentation at Day 1 of the Technical Workshop Series: Management of Pinniped Predation on Salmon at the Ballard Locks.
- Scordino, J.J. 2023. Update from Washington Academy of Sciences: Pinniped Predation on Salmonids. Presentation at Day 1 of the Technical Workshop Series: Management of Pinniped Predation on Salmon at the Ballard Locks.
- Scordino, J., and R. Pfeifer. 1993. Sea lion/steelhead conflict at the Ballard Locks: A history of control efforts to date and a bibliography of technical reports. NMFS-NWR Report.
- Stansell, R.J. 2004. Evaluation of pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2002-2004. USACE Report.
- Stansell, R.J. 2005. Summary of pinniped observations at Bonneville Dam in 2005. USACE Letter Report.
- Stansell, R.J. 2006. Summary of pinniped observations at Bonneville Dam in 2006. USACE Letter Report.
- Stansell, R.J., K.M. Gibbons, and W.T. Nagy. 2010. Evaluation of pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2008-2010. USACE Report.
- Stansell, R.J., K.M. Gibbons, W.T. Nagy, and B.K. van der Leeuw. 2011. Evaluation of pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2011. USACE Report.
- Stansell, R.J., K.M. Gibbons, W.T. Nagy, and B.K. van der Leeuw. 2012. 2012 Field Report: Evaluation of pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2012. USACE Report.
- Stansell, R.J., B.K. van der Leeuw, K.M. Gibbons, and W.T. Nagy. 2013. 2013 field report: Evaluation of pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2013. USACE Report.
- Stansell, R.J., B.K. van der Leeuw, K.M. Gibbons, and W.T. Nagy. 2014. Evaluation of Pinniped Predation on Adult Salmonids and Other Fish in The Bonneville Dam Tailrace, 2014. USACE Report.
- Steingass, S., B. Wright, S. Valentine, D. Heiner, S. Riemer, Z. Kroneberger, E. Nass, B. Sorenson, B. Triplet, J. Burco, and C. Gillin. 2019. Willamette Falls Management Report. ODFW Report.

- Steingass, S., S. Pearson, D. Hatch, and J. Dupont. 2020. Annual Report: 2020 Columbia River Basin Research and Management Activities. Report to NOAA/NMFS.
- Tackley, S.C., R.J. Stansell, and K.M. Gibbons. 2008. 2008 field report: evaluation of pinniped predation on adult salmonids and other fishes in the Bonneville Dam tailrace. USACE Field Report.
- Tackley, S.C., R.J. Stansell, and K.M. Gibbons. 2008. Pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2005-2007. USACE Report.
- Tidwell, K.S. 2023. Lessons learned from Bonneville - USACE perspective. Presentation at Day 2 of the Technical Workshop Series: Management of Pinniped Predation on Salmon at the Ballard Locks.
- Tidwell, K.S., and B.K. van der Leeuw. 2021. Evaluation of Pinniped Predation on Adult Salmonids and Other Fish in the Bonneville Dam Tailrace, 2020. USACE Report.
- Tidwell, K.S., B.K. van der Leeuw, L.N. Magill, B.A. Carrothers, and R.H. Wertheimer. 2017. Evaluation of pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2017 USACE Report.
- Tidwell, K.S., B.A. Carrothers, K.N. Bayley, L.N. Magill, and B.K. van der Leeuw. 2019. Evaluation of pinniped predation on adult salmonids and other fish in the Bonneville Dam tailrace, 2018. U.S. Army Corps of Engineers, Portland District, Fisheries Field Unit. Cascade Locks, OR. 65pp.
- Tidwell, K.S., B.A. Carrothers, D.T. Blumstein, and Z.A. Schakner. 2021. Steller Sea Lion (*Eumetopias jubatus*) Response to Non-lethal Hazing at Bonneville Dam. *Front. Conserv. Sci.* 2:760866.
- Tidwell, K.S., M.W. Braun, and B.K. van der Leeuw. 2022. Evaluation of Pinniped Predation on Adult Salmonids and Other Fish in the Bonneville Dam Tailrace, 2022. USACE Report.
- Urgenson, L., J. Kubo, and C. Degasperi. 2021. Temperature and Dissolved Oxygen Conditions in the Lake Washington Ship Canal and Impacts on Salmon. Prepared for the Lake Washington, Cedar, Sammamish Watershed (WRIA 8) Salmon Recovery Council.
- USACE (U.S. Army Corps of Engineers) Field Fish Unit. 2015. Evaluation of Pinniped Predation on Adult Salmonids and Other Fish in the Bonneville Dam Tailrace, 2015. USACE Report.
- van der Leeuw, B.K., and K.S. Tidwell. 2022. Evaluation of Pinniped Predation on Adult Salmonids and Other Fish in the Bonneville Dam Tailrace, 2021. USACE Report.
- Ward, D.L. 2002. Predator control program (mainstem/systemwide province). Program summary. Northwest Power Planning Council.
- WDFW (Washington Department of Fish and Wildlife). 2018. Spawning Escapement Update for the Lake Washington Watershed. WDFW PowerPoint presentation.
- Williams, R., E. Ashe, L. Bogaard, A. Bergman, T. Goetz, and V. Janik. 2021. Employing Targeted Acoustic Startle Technology (TAST) to deter harbor seal predation on endangered salmonids at the Ballard Locks, Seattle, WA. Oceans Initiative Report.
- Williams, R., E. Ashe, L. Bogaard, and A. Bergman. 2022. Acoustic Technology for Reducing Pinniped Predation on Salmon at the Ballard Locks. Oceans Initiative Report.

- Wright, B.E., S.J. Jeffries, R.F. Brown, R.J. Stansell, and D. Hatch. 2007. Field Report - Non-lethal pinniped deterrent activities at Bonneville Dam, Spring 2006. ODFW Report.
- Wright, B., T. Murtagh, and R. Brown. 2014. Willamette Falls Pinniped Monitoring Project. 2014 ODFW Report.
- Wright, B., T. Murtagh, and R. Brown. 2015. Willamette Falls Pinniped Monitoring Project. 2015 ODFW Report.
- Wright, B., T. Murtagh, and R. Brown. 2016. Willamette Falls Pinniped Monitoring Project. 2016 ODFW Report.
- Wright, B., T. Murtagh, R. Brown, and S. Reimer. 2017. Willamette Falls Pinniped Monitoring Project, 2017. ODFW Report.
- Wright, B., T. Murtagh, C. Owen, K. Sandoval, and T. Tillson. 2018. Willamette Falls Pinniped Monitoring 2018. ODFW Report.
- Wright, B., S. Steingass, C. Owen, E. Warren, M. Brown, S. Valentine, B. Triplett, and Z. Kroneberger. 2020. Annual Report: Pinniped Monitoring at Willamette Falls, 2019-2020. ODFW Report.
- Wright, B., S. Steingass, C. Owen, E. Warren, M. Brown, S. Valentine, B. Triplett, and B. Phibbs. 2021. Annual Report: Pinniped Monitoring at Willamette Falls, 2020-2021. ODFW Report.
- Wright, B., M. Brown, C. Owen, S. Valentine, Z. Kroneberger, E. Nass, and B. Phibbs. 2022. Annual Report: Pinniped Monitoring at Willamette Falls, 2021-2022. ODFW Report.
- WSAS (Washington State Academy of Sciences). 2022. Pinniped Predation on Salmonids in the Washington Portions of the Salish Sea and Outer Coast. Washington State Academy of Sciences 1-81.

APPENDIX A. DETAILED DESCRIPTION OF PINNIPED PREDATION CONTROL MEASURES USED AT IMPOUNDMENTS AND THEIR LIMITATIONS

A variety of measures have been considered, tested, and/or implemented to deter or remove pinnipeds from areas where their presence: (1) creates conflicts with other resources, (2) results in interactions with human activities, or (3) threatens human safety and/or damages property.

The following is a description and evaluation extracted from Scordino (2010) of the primary measures used by state and federal fishery/wildlife officials to control pinniped predation at impoundments where pinniped foraging is impacting the status of salmonid populations. The references cited in this appendix can be found in the above Reference section or in Scordino (2010).

The pinniped control measures are organized into five categories:

- 1) Noise stimuli
- 2) Vessel chasing or hazing
- 3) Tactile harassment
- 4) Capture/removal at or near the fishway
- 5) Operational/structural modifications at the impoundment site

Descriptions of other methods (such as predator sounds, predator [killer whale] models, taste aversion, electrical power [arc gap] discharge system, electric barriers, mid-range sonar, pinniped population reduction, etc.) that have been considered or tested and found to be ineffective or impractical can be found in Scordino (2010).

1. NOISE STIMULI

Like most other wildlife, pinnipeds react to noises in their environment, especially novel sounds that they are not commonly exposed to in their aquatic environment. The most commonly used noise stimuli to startle or cause pinnipeds to flee are handheld firecrackers that explode underwater (called firecrackers or seal cracker devices) or shotgun-dispatched firecrackers that explode on the surface of the water (called cracker shells). Acoustic devices that produce chirps or beeps underwater of varying frequencies (that can be heard by pinnipeds but not fish) and decibels are also commonly used or tested to deter pinnipeds from the area of transmission.

1.a. Firecrackers. State and federal wildlife managers routinely use firecrackers to scare wildlife. The commercially available underwater firecracker (called a seal control device or seal cracker device) consists of a spiral-wound cardboard tube containing 36 grains of potassium perchlorate and pyro-aluminum flash powder with an 8-second waterproof fuse and is weighted with sand so as to sink and explode underwater. They produce light and sound pressures on the order of 190 dB at 1 meter. Most of the sound energy is focused below 1.0 kHz, which is below the range of maximum hearing sensitivity for sea lions.

Firecrackers were initially successful in reducing sea lion predation on steelhead at the Ballard Locks. However, use of firecrackers subsequently became ineffective on several sea lions that habituated to preying on steelhead at the Ballard Locks (Pfeifer et al. 1989). Although these sea lions were initially frightened by the firecrackers, they soon began to either return in a few hours or resumed preying on steelhead in a different area. Some sea lions that had been observed over several seasons appeared to have learned to ignore or tolerate the noise. They also appeared to learn to evade close exposure to

firecrackers by diving and surfacing in unpredictable patterns (Pfeifer et al. 1989). No visible injuries to sea lions from firecrackers were observed during their use at the Ballard Locks (NMFS and WDFW 1995). Sea lions that were exposed to repeated use of firecrackers at the Ballard Locks from 1986 to 1988 were observed in subsequent years and continued to react to noise stimuli, indicating their hearing likely had not been affected long-term by their exposure to firecrackers.

1.b. Cracker Shells. Cracker shells are pyrotechnic devices discharged from a 12-gauge shotgun. The shells contain a flash explosive charge (same as a firecracker) that is designed to explode in the air or on the surface of the water at a distance of 75 to 100 meters from the point of discharge. The impulsive noise from the shotgun firing is comparable to firing a regular round of ammunition, and the noise from the cracker shell explosion is similar to a firecracker. Cracker shells were used to deter California sea lions at Bonneville Dam, with limited effectiveness in keeping sea lions from foraging on salmon (Brown et al. 2007).

1.c. Aerial Pyrotechnics. Aerial pyrotechnics (screamer rockets, poppers, banger rockets, bottle rockets) have been used to scare birds away from crops, to scare pinnipeds off docks, and to deter birds and sea lions at Bonneville Dam. The pyrotechnics are dispatched from a hand-held launcher (similar to a .22 short caliber starter pistol) and fly through the air, emitting a loud whistling sound (screamers) that end with “bang,” similar to a firecracker. The units were used at Bonneville Dam to reduce predation on juvenile salmonids, and their use was extended to keep sea lions away from the fish ladder at Bonneville Dam with limited success (NMFS 2008).

1.d. Acoustic Deterrents. Acoustic devices were developed in the 1980s to produce underwater noises at specific frequencies and with sufficient power to deter pinnipeds. Acoustic harassment devices (AHDs) were designed to produce high amplitude, pulsed but irregular sound underwater in the 12 to 17 kHz range that is intended to cause physical discomfort and to irritate pinnipeds, thereby repelling them from the area of the sound. The output of AHDs is designed to vary randomly to produce bursts of short, upswept tones at 160 dB to 185 dB.

AHDs were initially effective in some situations, but their effectiveness diminished in most situations as pinnipeds apparently learned to tolerate the noise. An AHD was used to attempt to control sea lion predation at the Ballard Locks, but was found to be ineffective (NMFS and WDFW 1995). The principal problem encountered with AHDs was that pinnipeds appeared to “learn” to tolerate the noise.

More powerful acoustic devices were later tested at the Ballard Locks. These more powerful devices, called acoustic deterrent devices (ADDs), were designed to cause pain to pinnipeds (rather than just a physical discomfort, irritation, or startling effect). Omni-directional ADDs (hung on the safety cable at Ballard) produced periodic sound emissions at 10 kHz with source levels that were measured to be between 190 and 196 dB at 1 meter. Directional ADDs (placed at the bottom of the spillway) were designed to be at or above the 200–220 dB estimated pain threshold for California sea lions. The omni and unidirectional arrays were placed in the spillway area at Ballard to create an “acoustic barrier” and “ensonified zone.”

The ADD acoustic barriers appeared to be effective in deterring newly arriving sea lions at the Ballard Locks area, but had less effect on California sea lions that repeatedly foraged at this site (NMFS 1996). However, even the “repeat” or habituated sea lions altered their behavior in the area of the ADDs. The repeat sea lions approaching the ADD array at the Ballard Locks did not usually enter the ensonified area

adjacent to the fish ladder, and on the few occasions when they did, they were there for very short periods of time. Their foraging behavior in the zone also was altered, with more time at the surface and a tendency to stay in areas of turbulence where the ADD signal would have been reduced. The propagation of the signal from the acoustic devices was strongly influenced by turbulence and entrained air caused by water spilling over the dam. Air bubbles in the water column absorb the acoustic signal, and sound levels decreased as spill increased.

ADDs also were used at Bonneville Dam. Directional transducers were placed on the bottom at the entrances to the fish ladders during the spring Chinook run. Observers did not report any obvious effects to the California sea lions foraging in the area (Tackley et al. 2008).

2. VESSEL CHASING OR HAZING

Boats have been used to attempt to scare or chase pinnipeds at the Ballard Locks and at Bonneville Dam. This method was not completely effective as pinnipeds in many cases simply swam under the boat and resisted leaving the area. Aggressive boat maneuvering combined with the use of firecrackers was initially effective at the Ballard Locks, but became less effective as California sea lions learned to avoid the boat or temporarily move downstream and then immediately return to the Ballard Locks (Pfeifer et al. 1989).

3. TACTILE HARASSMENT (shooting pinnipeds with rubber projectiles)

Shotgun-fired rubber buckshot and slugs designed to non-lethally repel bears have been used on California sea lions at Willamette Falls and Bonneville Dam. Rubber-tipped arrows shot from a crossbow were used on California sea lions at the Ballard Locks. The discharge of rubber projectiles was intended to deliver a non-lethal impact causing potential bruising but not penetrating the skin. The rubber projectiles were directed at the exposed part of the target animal's body, avoiding the head and eyes, to achieve the deterrent effect.

During use of rubber-tipped arrows at the Ballard Locks, some sea lions showed avoidance behavior after being hit while others did not. One sea lion appeared to avoid the area near the shooter, but still preyed on steelhead (Pfeifer et al. 1989).

At Willamette Falls, ODFW tested the use of rubber projectiles (rubber buckshot and batons) shot from a shotgun in 1986. The smaller and less commonly occurring sea lions immediately left the area when shot and did not return. Larger animals returned to the immediate area in less than 24 hours and were more wary on return, moving from area to area, surfacing in less predictable ways, and spending more time underwater, thus preventing dispatch of another rubber projectile (NMFS and ODFW 1997). The effect of paint balls dispatched from a CO₂ pistol at Willamette Falls was similar to the other deterrents in that the struck sea lions would move from the immediate area but continued foraging (Boatner 2000).

At Bonneville Dam, shotgun-dispatched rubber buckshot and batons were used on California sea lions in 2006 through 2008. Over 3,000 rubber buckshot/baton rounds were used from boats during the 3 years, with limited effectiveness in deterring sea lions (Brown et al. 2008).

Although most sea lions reacted when hit, they did not always leave the foraging area and in many instances immediately returned. Rubber projectiles also posed safety hazards to people in the immediate area due to potential ricochet, and thus could only be used in restricted areas.

4. CAPTURE/REMOVAL AT OR NEAR IMPOUNDMENT SITES

4.a. Capture on Traps and Relocation. Capture and relocation efforts with California sea lions at the Ballard Locks indicated that transporting captured sea lions relatively short distances (from Ballard to the outer Washington coast) was not an effective approach because the sea lions quickly returned. Similar results occurred with California sea lions relocated from Bonneville Dam and Willamette Falls to the outer Oregon coast. Longer distance relocation of California sea lions from the Ballard Locks to the Channel Islands off southern California also resulted in sea lions returning. This long-distance relocation did delay sea lion returns for at least 30 days, thereby providing a window of safe passage for migrating salmonids that season (NMFS and WDFW 1995). However, the disadvantage was that some of the targeted sea lions (those that had been captured/removed previously and returned to forage at the Ballard Locks) could not be recaptured (NMFS 1996). A harbor seal was also captured and relocated a relatively short distance (Ballard Locks to Hood Canal), and the seal soon returned to forage at the Ballard Locks.

4.b. Capture on Traps and Captive Holding. California sea lions at the Ballard Locks and Bonneville Dam have been captured and placed in captivity permanently to eliminate the conflicts they caused. Although permanent captivity does eliminate the problem sea lions without having to kill them, the method is limited by the availability and interests of display facilities. One of the California sea lions captured at the Ballard Locks was placed in temporary captivity and released after the steelhead run. Temporary holding was found to be ineffective in the long term because this sea lion returned the following season and could not be recaptured before it had again preyed on salmonids at the Ballard Locks (NMFS 1996).

4.c. Capture and Lethal Injection. California sea lions have been selectively captured at both Bonneville Dam and at Willamette Falls and transported off-site for lethal injection under Section 120 of the MMPA. This method did result in lowered presence and predation on salmon. More recently, Steller sea lions have also been lethally removed.

4.d. Capture in Open Waters and/or Shooting. This method has not been used.

5. OPERATIONAL/STRUCTURAL MODIFICATIONS AT THE IMPOUNDMENT SITE

5.a. Physical Barriers. At both Bonneville Dam and Willamette Falls, bar structures (called sea lion exclusion devices [SLEDs] or barrier gates) have been placed inside the entry to the fishway to prevent sea lions from entering the fish ladder. At Bonneville Dam, the exclusion devices were installed at the entrance to each fish ladder to prevent sea lions from entering the fish ladder (Tackley et al. 2008). The welded aluminum grate structures, consisting of a series of evenly spaced vertical bars, were installed in the eight fish ladder entrances just prior to the spring Chinook salmon migration each year. The bars provided sufficient spacing for migrating salmon to pass, but the spacing was too narrow for sea lions to easily enter. A similar grate structure was installed at the fish ladder entrance at Willamette Falls to prevent sea lions from entering the fish ladder (NMFS and ODFW 1997). Unfortunately, lack of access to inside the fish ladder did not deter sea lions from feeding on salmonids near the ladder entrance.

A physical barrier was tested at the Ballard Locks to prevent sea lion access to a prime forage area near the entrance to the fish ladder (sea lions were not entering the fish ladder, but foraged effectively on steelhead as they approached the ladder entrance). The experimental barrier at the Ballard Locks (a large-mesh net strung underwater) was ineffective because fish passage may have been hampered by

the barrier and because sea lions quickly learned to effectively forage on steelhead at the face of the barrier (NMFS and WDFW 1995).

5.b. Operational Changes. At the Ballard Locks, the opening rotation of the spill gates was modified such that the flow was greatest at the entrance to the fish ladder, thus improving attraction water to the ladder. This was intended to reduce the time that salmon spend in the spillway area where they are more vulnerable to pinniped predation.

APPENDIX B. FIGURES SHOWING AREA BETWEEN THE SALMON BAY RAILROAD BRIDGE AND THE BALLARD LOCKS

